

Wave Run-up and Overtopping on Sloping Structures

The Iribarren number, EurOtop (2018) mean run-up and overtopping formulae, roughness and obliquity factors, and tolerable-discharge limits. Companion note to the online Run-up & Overtopping calculator.

1. Iribarren number and breaker type

Run-up and overtopping on a slope are governed by the **Iribarren number** (surf-similarity parameter), the ratio of the structure steepness to the square root of the wave steepness. Using the spectral wave height H_{m0} and the spectral period $T_{m-1,0}$ ($\approx T_m/1.1$ for a standard JONSWAP spectrum):

$$\xi_{m-1,0} = \tan\alpha / \sqrt{s_{m-1,0}}, \quad s_{m-1,0} = H_{m0}/L_{m-1,0}, \quad L_{m-1,0} = g \cdot T_{m-1,0}^2 / 2\pi.$$

The Iribarren number also classifies the breaker on the slope: **spilling** ($\xi < 0.5$), **plunging** ($0.5 < \xi < 3.3$) and **surging / collapsing** ($\xi > 3.3$). Run-up increases with ξ and then plateaus for surging waves.

2. Run-up: $R_{2\%}$

The 2% run-up height (exceeded by 2% of incident waves) on a straight impermeable smooth slope, after EurOtop (2018), is the smaller of a rising branch and a plateau:

$$R_{2\%}/H_{m0} = 1.65 \cdot \gamma_f \cdot \gamma_\beta \cdot \xi_{m-1,0}, \quad \text{with maximum } R_{2\%}/H_{m0} = 1.00 \cdot \gamma_f \cdot \gamma_\beta \cdot (4.0 - 1.5/\xi_{m-1,0}).$$

The classic **Hunt (1959)** formula $R = H \cdot \xi$ is reported alongside for comparison. The influence factors γ_f (roughness) and γ_β (obliquity) reduce run-up below the smooth, head-on value.

3. Mean overtopping discharge: q

For the mean overtopping rate per metre of crest with freeboard R_c , EurOtop (2018) gives (again taking the smaller of two branches):

$$q/\sqrt{(g \cdot H_{m0}^3)} = (0.023/\sqrt{\tan\alpha}) \cdot \gamma_h \cdot \xi \cdot \exp[-(2.7 \cdot R_c / (H_{m0} \cdot \xi \cdot \gamma_h \gamma_f \gamma_\beta))^{1.3}]$$

$$q/\sqrt{(g \cdot H_{m0}^3)} = 0.09 \cdot \exp[-(1.5 \cdot R_c / (H_{m0} \cdot \gamma_f \gamma_\beta))^{1.3}] \quad (\text{maximum}).$$

Overtopping falls very steeply as the crest freeboard rises, so R_c is the primary design lever. Obliquity uses $\gamma_\beta = 1 - 0.0033|\beta|$ for overtopping and $1 - 0.0022|\beta|$ for run-up ($|\beta| \leq 80^\circ$).

4. Design curves

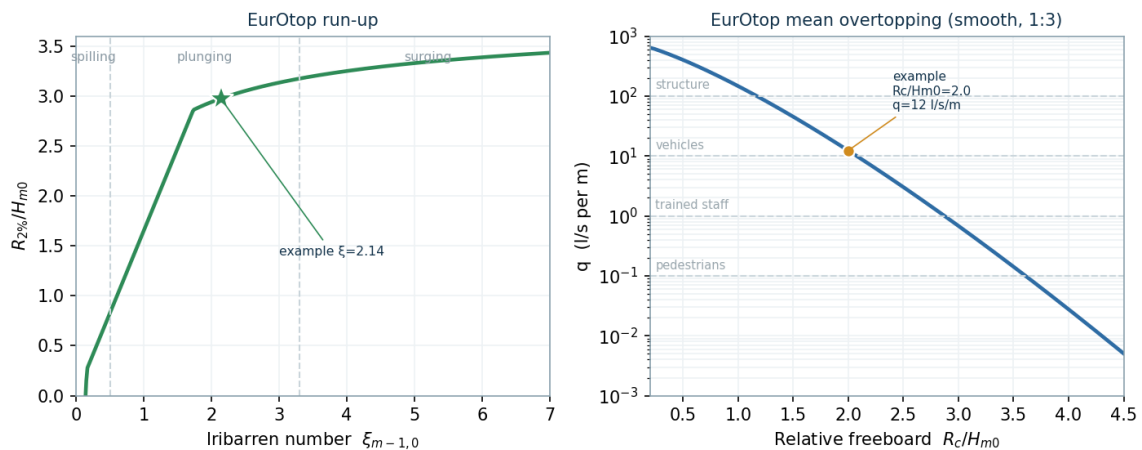


Figure 1. Left: EurOtop run-up $R_{2\%}/H_{m0}$ versus Iribarren number. Right: mean overtopping q versus relative freeboard R_c/H_{m0} (smooth 1:3 slope), with tolerable-limit guides. Markers show the worked example of Section 6.

5. Roughness factors and tolerable limits

Typical surface roughness factors γ_f (EurOtop 2018): smooth concrete/asphalt 1.00; grass 1.00; rock, one layer on impermeable core 0.60; rock, two layers on a permeable core 0.40; cubes (2 layers) 0.47; Accropode/Core-loc/Xbloc 0.46; tetrapods 0.38. Rough armour therefore cuts run-up and overtopping sharply. Indicative tolerable mean discharges:

Receptor / hazard	Tolerable mean q (l/s per m)
Pedestrians, unaware / poor view	0.03
Pedestrians, aware ($H_{m0} = 1\text{--}3$ m)	0.1 – 1

Trained staff, well shod	1 – 10
Vehicles driving at low speed	~10 – 50
Embankment / seawall — damage to grass or unpaved crest	5 – 50
Reinforced / armoured crest and rear	50 – 200

6. Worked example

Quantity	Value
Inputs	$H_{m0} = 2.0$ m, $T_p = 8.0$ s, $\cot\alpha = 3$, $R_c = 4.0$ m, $\gamma_f = 1.0$, $\beta = 0^\circ$
Spectral period, $T_{m-1,0} = T_p / 1.1$	7.27 s
Deep-water length, $L_{m-1,0}$	82.6 m
Wave steepness, $s_{m-1,0}$	0.0242
Iribarren number, $\xi_{m-1,0}$	2.14 (plunging)
2% run-up, $R_{2\%}$ (EurOtop)	5.95 m
Run-up, R (Hunt)	4.28 m
Mean overtopping, q	12.3 l/s per m
Crest check	$R_{2\%} = 5.95$ m $>$ $R_c = 4.0$ m $\rightarrow$ green-water overtopping
Assessment	Dangerous — at the vehicle / structure tolerable limit

Notation

H_{m0} spectral significant wave height at the toe; T_p peak period; $T_{m-1,0}$ spectral (energy) period; $L_{m-1,0}$ deep-water length; α slope angle ($\cot\alpha$ = horizontal:vertical); ξ Iribarren number; $R_{2\%}$ 2% run-up height; R_c crest freeboard above still water; q mean overtopping discharge; γ_f roughness factor; γ_β obliquity factor; γ_b berm factor (= 1 here).

References

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